Pseudohypoparathyroidism Presenting as Symptomatic Hypocalcaemia with Tetany in a 43-Year-Old Tanzanian Woman: A Case Report and Systematic Literature Review

James Masanja,[1] Silas Gamba,[1] Denis Rainer[1]

[1]Department of Internal Medicine, Muhimbili National Hospital, Dar es Salaam, Tanzania

Corresponding authors: James Masanja, Silas Gamba, Denis Rainer

ABSTRACT

Pseudohypoparathyroidism (PHP) is a rare hereditary endocrine disorder characterised by target-organ resistance to parathyroid hormone (PTH), resulting in hypocalcaemia, hyperphosphataemia, and elevated serum PTH levels despite normal renal function. We report a case of a 43-year-old Tanzanian woman presenting with a five-month history of progressive peri-oral and acral paraesthesia, followed by increasingly severe carpopedal, jaw, and generalized tetanic muscle spasms. Physical examination revealed a positive Trousseau sign and symmetrical hyperreflexia. Investigations demonstrated severe ionised hypocalcaemia (iCa 0.43 mmol/L; reference 1.12–1.32 mmol/L), hyperphosphataemia (1.6 mmol/L), and markedly elevated PTH (176.8 pg/mL; reference 6–80 pg/mL) with preserved renal function, establishing a biochemical diagnosis of PTH resistance. After exclusion of hypoparathyroidism, vitamin D deficiency, chronic kidney disease, and malabsorption syndromes, a clinical diagnosis of pseudohypoparathyroidism was made. The patient was managed with intravenous calcium gluconate, followed by oral calcitriol and calcium carbonate, with clinical improvement. This case highlights the importance of recognising PHP as a differential diagnosis for hypocalcaemia and tetany, particularly in resource-limited settings where genetic testing remains unavailable, and underscores the critical gap in epidemiological data on PHP in sub-Saharan Africa.

Keywords: Pseudohypoparathyroidism; Hypocalcaemia; Tetany; PTH resistance; GNAS; Sub-Saharan Africa; Tanzania

INTRODUCTION

Pseudohypoparathyroidism (PHP) is a heterogeneous group of rare endocrine disorders united by resistance of target organs — primarily the kidney and bone — to the actions of parathyroid hormone (PTH), despite the production of biologically intact hormone in supranormal quantities.1 The biochemical hallmark is a triad of hypocalcaemia, hyperphosphataemia, and elevated serum PTH in the presence of normal renal function.2 First described by Fuller Albright and colleagues in 1942, PHP remains one of the most complex endocrine disorders in clinical practice, characterised by molecular heterogeneity involving the GNAS locus on chromosome 20q13.32, which encodes the stimulatory G-protein alpha subunit (Gsα) crucial for PTH receptor signal transduction.3

The clinical spectrum of PHP is broad, ranging from asymptomatic biochemical abnormalities to life-threatening hypocalcaemic crises manifesting as tetany, laryngospasm, seizures, and cardiac arrhythmias.4 Some subtypes, particularly PHP Type Ia and Ib, may present with phenotypic features of Albright Hereditary Osteodystrophy (AHO), including short stature, obesity, round face, brachydactyly, and subcutaneous ossifications.5 However, PHP Type Ic and II present with PTH resistance without AHO features, creating diagnostic challenges that frequently result in delayed or missed diagnoses.6

Globally, the estimated prevalence of PHP ranges from 0.62 to 1.47 cases per 100,000 population.7 The condition occurs approximately twice as frequently in females as in males, and symptoms most commonly emerge during childhood and adolescence, although adult-onset presentations are well-documented.7 Critically, no epidemiological data exist for sub-Saharan Africa or Tanzania, and published case reports from this region are absent from the indexed literature, creating a significant knowledge gap that may reflect underdiagnosis rather than true rarity.8

This case is of particular academic and clinical significance for several reasons: it represents the first documented case report of PHP from Tanzania; it illustrates the diagnostic challenges in resource-limited settings where genetic testing, DEXA scanning, and molecular analysis are unavailable; and it demonstrates that PHP can present in middle adulthood without classical AHO features, necessitating a high index of clinical suspicion. We present this case with a comprehensive literature review to raise awareness among clinicians in East Africa and to highlight the need for systematic epidemiological research in this region.

CASE PRESENTATION

Patient Demographics and History of Presenting Illness

A 43-year-old Tanzanian woman (H.H.R.), a ~~small-scale businesswoman from Tabora with primary-level education, presented to Muhimbili National Hospital (MNH) on 7 May 2026, referred from Kitete Regional Referral Hospital.~~ She had no known prior comorbidities, no history of surgery, blood transfusion, or medication use, and denied alcohol or tobacco consumption.

Her presenting complaint comprised two sequential symptoms. Approximately five months prior to admission (early December 2025), she developed a gradual-onset tingling sensation described as "pinpricks and needles" with associated numbness, initially confined to the peri-oral region and subsequently spreading to involve both hands and feet bilaterally. The sensations lacked

clear periodicity but were noted to worsen with anxiety. During the preceding month, symptoms had intensified, causing difficulty holding objects and walking.

Four months prior to admission (~~January 2026~~), she developed additional episodic muscle spasms, initially presenting as intermittent twitching of the lips and eyelids bilaterally, occurring two to three times per week, each episode lasting five to ten seconds, and without pain. Over the subsequent three months, the spasms progressively worsened in frequency (four to six times per week), duration (three to five minutes), and intensity, with involvement of the hands and feet causing digital flexion and plantar curling, accompanied by pain (Visual Analogue Scale [VAS] 4–5/10).

One week prior to admission, spasm frequency escalated to three to five episodes per day with severe pain (VAS 6–8/10). On the day before admission, she developed recurrent episodes of jaw stiffness and generalised painful muscle spasms, each lasting approximately fifteen minutes, associated with inability to speak, difficulty feeding, inability to walk, palpitations, and heightened anxiety.

A systematic review of systems was negative for features of thyroid dysfunction, chronic kidney disease, malabsorption, connective tissue disease, haematological disorder, or prior open wounds. There was no history of neck surgery, radiation, or use of medications known to cause hypocalcaemia (diuretics, proton pump inhibitors, anticonvulsants, bisphosphonates). Dietary assessment revealed adequate caloric and nutritional intake, including regular dairy, protein, and fruit consumption.

Past Medical, Obstetric, and Family History

The patient had one prior hospital admission in 2018 for malaria. Her obstetric history revealed parity of 2 (both normal vaginal deliveries in 1998 and 2002) with no obstetric complications, complete tetanus vaccination, and current IUCD use. Menarche occurred at 13 years with regular menstrual cycles. Family history was unremarkable, with no similar presentations among nine siblings or other relatives.

Physical Examination

On examination, the patient was conscious, alert, and oriented (GCS 15/15). She was obese (weight 87 kg, height 163 cm, BMI 32.7 kg/m², waist circumference 82 cm) without dyspnoea, pallor, cyanosis, jaundice, lymphadenopathy, or peripheral oedema. Vital signs revealed blood pressure 115/71 mmHg, pulse rate 84 bpm (regular), respiratory rate 19 breaths/min, SpO_2 98% on room air, temperature 36.7°C, and random blood glucose 6.6 mmol/L.

Neurological examination demonstrated normal higher cortical function, intact cranial nerves (including a ~~negative~~ Chvostek sign at the facial nerve), and no meningeal signs. Motor examination revealed symmetrical brisk deep tendon reflexes (3+) in both upper and lower limbs. Sensory, coordination, and gait examinations were entirely normal. Significantly, Trousseau's sign was positive on occlusion of the upper arm with a blood pressure cuff (Figure 1 and Figure 2), and Archibald's metacarpal sign was negative.

Thyroid examination revealed no visible swelling, palpable nodules, or tenderness. Cardiovascular, respiratory, abdominal, breast, and integumentary examinations were all within normal limits. Digital rectal examination was unremarkable, and urine dipstick was normal.





Figure 1 (left): Positive Trousseau sign elicited by sphygmomanometer cuff occlusion of the right upper arm, demonstrating carpopedal spasm. Figure 2 (right): Bilateral flexion of the fingers towards the palm observed during the episode of hypocalcaemic tetany.

Investigations

Table 1 summarises the key biochemical findings. Arterial blood gas analysis confirmed normal pH (7.41), pCO_2 (36.7 mmHg), pO_2 (81 mmHg), and bicarbonate (25 mmol/L), excluding respiratory alkalosis as a cause of tetany. Ionised calcium on ABG was critically low at 0.43 mmol/L (reference 1.12–1.32 mmol/L). Serum calcium on formal biochemistry was 1.9 mmol/L (reference 2.1–2.55 mmol/L), with elevated phosphorus at 1.6 mmol/L (reference 0.81–1.45 mmol/L) and normal magnesium (1.03 mmol/L).

The decisive diagnostic finding was a markedly elevated intact PTH of 176.8 pg/mL (reference 6–80 pg/mL) in the context of hypocalcaemia and hyperphosphataemia with completely normal renal function (creatinine 53.7 µmol/L, BUN 2.8 mmol/L, eGFR normal). Vitamin D (25-OH) was mildly low at 17.7 ng/mL but not in the deficiency range typically associated with secondary hyperparathyroidism causing symptomatic hypocalcaemia. Thyroid function (TSH 1.546 mIU/mL, FT3 2.16 pg/mL, FT4 1.40 pg/mL), liver function, coagulation, serum albumin, and reproductive hormones were all within normal reference ranges.

Lipid profile revealed elevated total cholesterol (6.08 mmol/L), elevated LDL-cholesterol (5.04 mmol/L), and reduced HDL-cholesterol (0.9 mmol/L), consistent with dyslipidaemia in the context of obesity. Full blood count showed a mild normocytic normochromic anaemia (Hgb 12.14 g/dL) on admission, resolving to 11.2 g/dL at follow-up, with a reactive leucocytosis on day 1 (WBC 11.5 $\times 10^9$/L) that normalised. Serology was negative for HIV, HBsAg, and HCV antibody. Serum glucose and HbA1c were normal.

Electrocardiogram (Figure 3) demonstrated sinus tachycardia (rate 109 bpm, PR 126 ms, QT 338 ms, QTc 457 ms). The mildly prolonged QTc is a recognised feature of hypocalcaemia.9 Echocardiogram (Figure 4) was entirely normal. Kidney-ureter-bladder ultrasound (Figure 5)

demonstrated normal bilateral kidneys and no evidence of nephrocalcinosis or urolithiasis. Brain CT scan (Figure 6) revealed bilateral symmetrical basal ganglia calcification, a classic radiological feature of long-standing PHP.10

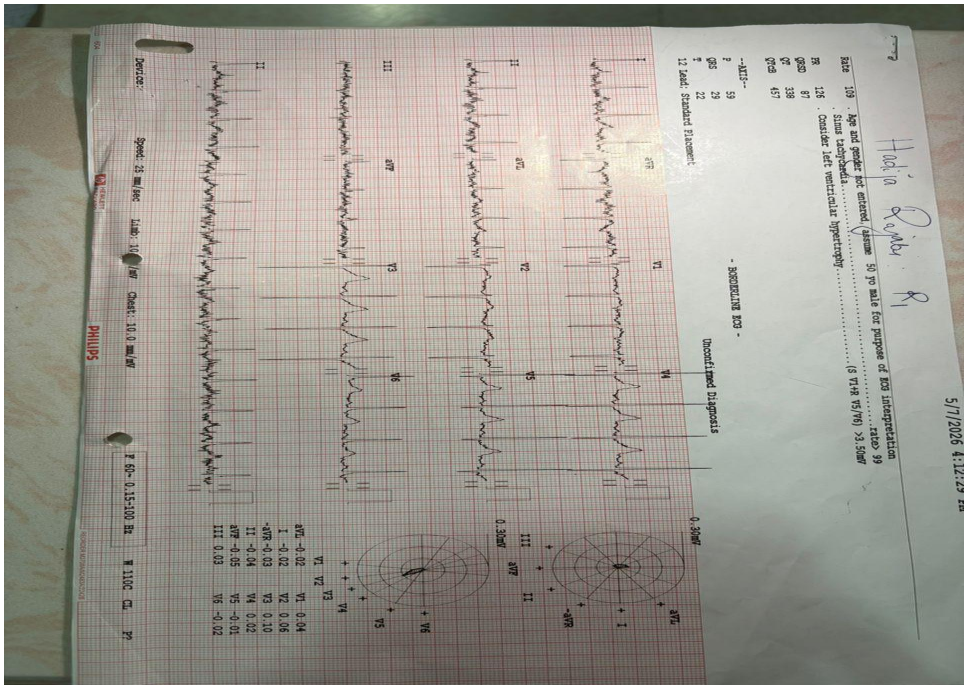


Figure 3: 12-lead ECG (7 May 2026) demonstrating sinus tachycardia (rate 109 bpm) with a prolonged QTc of 457 ms, consistent with hypocalcaemia-induced QT prolongation.

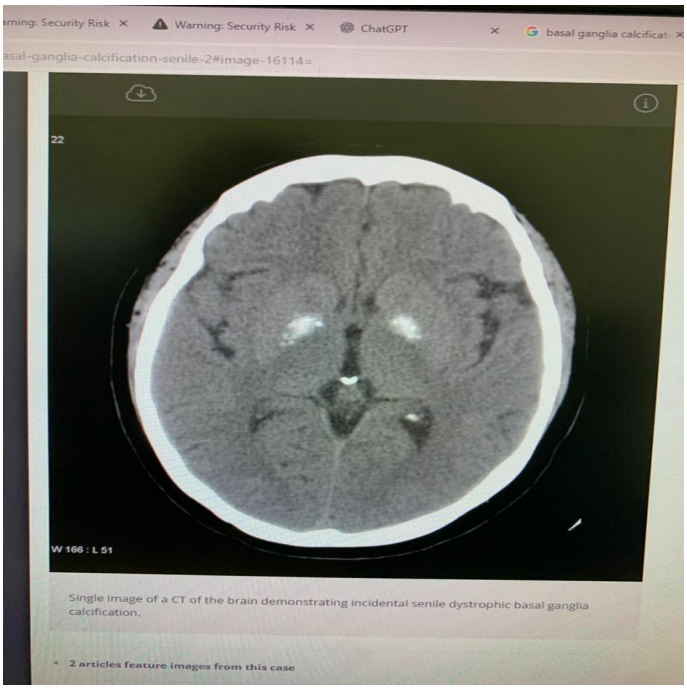


Figure 4: CT brain demonstrating bilateral symmetrical basal ganglia calcification (bilateral lenticular hyperdensities), a hallmark finding in long-standing pseudohypoparathyroidism.

Table 1: Summary of Key Biochemical Investigations

Parameter	Result	Reference Range	Interpretation
Ionised Calcium (ABG)	0.43 mmol/L	1.12–1.32 mmol/L	Severely low
Serum Calcium (total)	1.9 mmol/L	2.1–2.55 mmol/L	Low
Serum Phosphorus	1.6 mmol/L	0.81–1.45 mmol/L	Elevated
Intact PTH	176.8 pg/mL	6–80 pg/mL	Markedly elevated
Serum Magnesium	1.03 mmol/L	0.66–1.07 mmol/L	Normal
Creatinine	53.7 µmol/L	63.6–110.5 µmol/L	Normal (low-normal)
25-OH Vitamin D	17.7 ng/mL	6.6–49.9 ng/mL	Insufficiency
Serum Albumin	36.1 g/L	35.6–50.4 g/L	Normal
TSH	1.546 mIU/mL	0.49–4.67 mIU/mL	Normal
Total Cholesterol	6.08 mmol/L	<5.17 mmol/L	Elevated
LDL-Cholesterol	5.04 mmol/L	<3.34 mmol/L	Elevated
QTc (ECG)	457 ms	<440 ms	Borderline prolonged

Table 1: Key biochemical investigations demonstrating the classic biochemical triad of PHP — hypocalcaemia, hyperphosphataemia, and markedly elevated PTH — with preserved renal function.

Table 2: Complete Metabolic and Haematological Panel

Test	Value	Reference Range	Status
Sodium (mmol/L)	139	135–145	Normal
Potassium (mmol/L)	4.0	3.5–5.1	Normal
Chloride (mmol/L)	99	98–107	Normal
BUN (mmol/L)	2.8	2.5–7.4	Normal
Uric Acid (mmol/L)	0.31	0.15–0.35	Normal
ALP (U/L)	82.53	45–155	Normal
AST (U/L)	19.5	0–48.1	Normal
ALT (U/L)	13.0	0–55	Normal
Total Bilirubin (µmol/L)	10.2	5.2–41	Normal
Haemoglobin (g/dL)	12.14	12–15	Low-normal
WBC ($\times 10^9$/L)	11.5 → 6.5	4–10	Reactive → Normal
Platelets ($\times 10^9$/L)	360.7	150–410	Normal
HbA1c (%)	5.5	4–6	Normal
Vitamin B12 (pg/mL)	605	187–883	Normal
Folate (ng/mL)	8.3	7.2–15.4	Normal

Table 2: Comprehensive metabolic and haematological results, demonstrating absence of renal failure, liver disease, haematological malignancy, or nutritional deficiency as alternative explanations for PTH elevation.

Differential Diagnosis

The diagnostic work-up systematically addressed four principal differential diagnoses for the biochemical triad of hypocalcaemia, hyperphosphataemia, and elevated PTH, as summarised in Table 3.

Table 3: Differential Diagnosis and Supporting/Refuting Evidence

Diagnosis	Features Supporting	Features Against
Hypoparathyroidism	Hypocalcaemia, tetany, carpopedal spasm, paraesthesia, hyperreflexia	PTH markedly elevated (not low/absent); no neck surgery, autoimmune disease, DiGeorge syndrome, infiltrative disease, or radiation history
Secondary Hyperparathyroidism	Elevated PTH, hypocalcaemia symptoms	Normal renal function; no malabsorption, chronic diarrhoea, or vitamin D deficiency of sufficient severity; adequate diet
Pseudohypoparathyroidism (PHP)	Hypocalcaemia + hyperphosphataemia + markedly elevated PTH with normal renal function; obesity; positive Trousseau sign; bilateral basal ganglia calcification on CT brain	No classical AHO phenotype (no short stature, brachydactyly, round face, subcutaneous ossifications); no family history; adult onset — consistent with PHP Ic or II variant
Hypomagnesaemia	May cause tetany and paraesthesia; can impair PTH secretion	Serum magnesium normal (1.03 mmol/L); no alcohol use, diuretics, or chronic diarrhoea

Table 3: Systematic approach to differential diagnosis for symptomatic hypocalcaemia with elevated PTH and normal renal function.

Diagnosis

Based on the First International Consensus Statement diagnostic criteria for PHP (Mantovani et al., 2018), a diagnosis of pseudohypoparathyroidism was established on the basis of: (1) hypocalcaemia (serum Ca 1.9 mmol/L, iCa 0.43 mmol/L); (2) hyperphosphataemia (1.6 mmol/L); (3) markedly elevated intact PTH (176.8 pg/mL); and (4) normal renal function, excluding secondary hyperparathyroidism due to renal failure.11 The absence of classical AHO features (short stature, brachydactyly, subcutaneous ossifications, round face) suggests PHP Type Ic or Type II, though definitive molecular subtyping would require GNAS gene analysis, which was not available. Bilateral basal ganglia calcification on CT brain provided additional supportive evidence.

Management

Acute management was initiated in the emergency department with intravenous calcium gluconate 2 g administered in repeated cycles until symptomatic improvement was achieved. Inpatient management was then transitioned to oral calcitriol 0.25 mcg once daily and calcium carbonate 500 mg twice daily, with close monitoring of serum calcium and phosphorus. The patient was counselled regarding lifestyle modification (dietary calcium optimisation, vitamin D supplementation, sunlight exposure) and the hereditary nature of the condition. Genetic

counselling was initiated given the implications for family members, and genetic testing was listed as a priority upon availability.

Lipid-lowering therapy for the identified dyslipidaemia was deferred pending outpatient review, given the acute metabolic priority. Follow-up was arranged at two weeks for repeat electrolytes, with a planned endocrinology outpatient clinic review at four weeks.

Outcome and Follow-Up

Following initiation of intravenous calcium gluconate, the patient demonstrated prompt clinical improvement with resolution of jaw stiffness and generalised spasms within 24 hours. Transition to oral therapy was well-tolerated. The patient was discharged in a stable condition with outpatient follow-up appointments. Prognosis is considered good with appropriate long-term calcium and calcitriol supplementation, though lifelong treatment and monitoring are required. The patient was advised regarding potential complications of prolonged hypocalcaemia, including cataracts, dental abnormalities, and neuropsychiatric manifestations.

DISCUSSION AND COMPREHENSIVE LITERATURE REVIEW

Pathophysiology of Pseudohypoparathyroidism

PTH is secreted by the chief cells of the parathyroid glands in response to hypocalcaemia and acts through the PTH/PTHrP receptor (PTH1R) — a class B G-protein coupled receptor — on renal tubular and osteoblastic cells.12 The downstream signalling cascade requires coupling of PTH1R to the stimulatory G-protein alpha subunit (Gsα), encoded by the GNAS gene on chromosome 20q13.32, which activates adenylyl cyclase to generate cyclic AMP (cAMP) as the principal intracellular second messenger.3 In PHP, defects at the level of the receptor or downstream signalling prevent appropriate target organ response, resulting in persistent hypocalcaemia and hyperphosphataemia despite supraphysiological PTH concentrations.13

The GNAS locus is subject to tissue-specific genomic imprinting. Gsα is expressed biallelically in most tissues, but predominantly from the maternal allele in renal proximal tubules, pituitary, thyroid, and gonads.14 This imprinting explains the parent-of-origin effects observed in PHP: maternally-inherited GNAS mutations cause PHP Type Ia (with full PTH and other hormonal resistance), whereas paternally-inherited mutations cause pseudo-pseudohypoparathyroidism (PPHP) — AHO features without biochemical resistance.5

Classification of Pseudohypoparathyroidism

PHP is classified into multiple subtypes based on molecular aetiology, inheritance pattern, biochemical findings, and clinical features. The most widely used classification system is summarised in Table 4, based on the First International Consensus (2018) and the molecular framework proposed by Jüppner (2021).1,3

Table 4: Classification of Pseudohypoparathyroidism and Related Disorders

Type	Gene/Locus	Molecular Defect	Inheritance	Biochemistry	AHO Features
PHP-Ia	GNAS (20q13.32)	Inactivating Gsα mutation (maternal allele)	AD (maternal)	Hypocalcaemia + hyperphosphataemia + ↑PTH + multi-hormonal resistance	Present (short stature, obesity, brachydactyly, round face)
PHP-Ib	GNAS (epigenetic)	Imprinting defect of GNAS exon A/B	AD (maternal imprinting)	Hypocalcaemia + hyperphosphataemia + ↑PTH (isolated or mild)	Often absent or mild
PHP-Ic	GNAS (20q13.32)	Truncating Gsα mutation	AD (maternal)	Hypocalcaemia + hyperphosphataemia + ↑PTH	Absent
PHP-II	PTH1R (3p22.1)	PTH receptor defect	AD	Hypocalcaemia + hyperphosphataemia + ↑PTH	Absent

PPHP	GNAS (paternal allele)	Inactivating Gsα mutation (paternal allele)	AD (paternal)	Normal calcium/PTH	Present

Table 4: Molecular classification of pseudohypoparathyroidism and related disorders based on First International Consensus (2018) and Jüppner (2021). AD = Autosomal Dominant; PHP = Pseudohypoparathyroidism; PPHP = Pseudo-pseudohypoparathyroidism.1,3

Epidemiology

PHP is among the rarest endocrine conditions encountered in clinical practice. A landmark population-based study from France (Thiele et al., 2015) estimated a prevalence of 1.1 per 100,000 individuals, with PHP-Ib accounting for the largest proportion of cases.15 A similar Italian study reported a prevalence of 0.93 per 100,000.16 A systematic review published in 2021 reported pooled global prevalence estimates ranging from 0.62 to 1.47 per 100,000, with a consistent female predominance (F:M ratio approximately 2:1).7 The onset of symptoms most commonly occurs in childhood (median age 7–8 years for PHP-Ia) or adolescence.17

Critically, no published prevalence or incidence data exist for sub-Saharan Africa, East Africa, or Tanzania. This knowledge gap may partly reflect the logistical and financial barriers to biochemical and genetic diagnosis in these settings, including limited access to PTH assays, intact PTH immunoassays, and molecular genetics laboratories. The present case suggests that PHP is likely underdiagnosed in Tanzania, and systematic surveillance studies are urgently needed.

Clinical Manifestations and Diagnostic Challenges

The clinical presentation of PHP is protean and depends heavily on the subtype, the degree of hypocalcaemia, and the presence or absence of AHO features.18 Neuromuscular manifestations of hypocalcaemia — including perioral and acral paraesthesia, Chvostek sign, Trousseau sign, carpopedal spasm, laryngospasm, bronchospasm, and generalised tetany — dominate the acute presentation, as illustrated in the current case.4 Seizures may occur, particularly in children or in the context of severe biochemical derangement.19

PHP Type Ia presents with a full constellation of features: PTH resistance, AHO phenotype (short stature, obesity, round face, brachydactyly involving the 4th and 5th metacarpals/metatarsals, subcutaneous ossifications), and variable resistance to other Gsα-coupled hormones (TSH, LH/FSH, GHRH, glucagon), which may manifest as subclinical hypothyroidism, hypogonadism, or growth hormone deficiency.5 The current patient demonstrated obesity consistent with AHO but lacked short stature, brachydactyly, round facies, or subcutaneous ossifications, suggesting PHP-Ic or PHP-II rather than PHP-Ia. Normal thyroid function tests and reproductive hormones in our patient further argue against multi-hormonal resistance.

PHP-Ib typically presents with isolated PTH resistance without AHO features, caused by epigenetic defects at the GNAS locus.20 PHP-II is distinguished biochemically by a normal urinary cAMP response to PTH infusion (Ellsworth-Howard test) despite hypocalcaemia,

suggesting a post-cAMP defect in the signalling cascade.6 Without the Ellsworth-Howard test — unavailable in our setting — definitive subtype classification was not possible.

Long-term untreated or undertreated PHP carries significant morbidity, including posterior subcapsular cataracts (reported in up to 50% of adults with chronic hypocalcaemia), dental enamel defects and hypodontia, basal ganglia and cerebellar calcification (detected in our patient on CT brain), neuropsychiatric manifestations (depression, anxiety, cognitive impairment), cardiac arrhythmias from QT prolongation, and exacerbation of metabolic syndrome.21,22 The QTc of 457 ms observed in our patient reflects calcium's key role in cardiac repolarisation.

Diagnosis

The diagnosis of PHP rests on demonstrating the biochemical triad of PTH resistance: hypocalcaemia, hyperphosphataemia, and elevated serum PTH in the absence of renal failure, as codified by the First International Consensus Statement (Mantovani et al., 2018).11 All four biochemical criteria were clearly met in our patient. Albumin-corrected or ionised calcium should be used to avoid misclassification due to hypoalbuminaemia; in this patient, serum albumin was normal and both uncorrected and ionised calcium were low.

Key investigations for suspected PHP include: serum calcium (albumin-corrected or ionised), phosphate, PTH (intact), magnesium, creatinine (with eGFR), 25-OH vitamin D, urinary cAMP (for subtype differentiation), skeletal radiographs (for AHO features and heterotopic ossifications), CT brain (for basal ganglia calcifications), and GNAS genetic/epigenetic analysis (for definitive subtyping).23 The diagnostic algorithm proposed by the 2023 European Reference Network (ERN) guidelines recommends that clinicians consider PHP in any patient with unexplained hypocalcaemia and hyperphosphataemia once renal failure and hypoparathyroidism are excluded, regardless of the presence or absence of AHO features.24

Management

The management of PHP is directed towards normalising serum calcium and preventing end-organ complications of chronic hypocalcaemia and hyperphosphataemia.25 Unlike hypoparathyroidism — where replacement PTH (rhPTH 1–84, teriparatide) is now an established option — the PTH resistance in PHP means that PTH analogues have a limited role, and treatment centres on calcium and calcitriol supplementation.26

Acute hypocalcaemic crisis, as in our patient, requires intravenous calcium gluconate (1–2 g over 10–20 minutes, repeated as necessary) with continuous cardiac monitoring given the risk of arrhythmias.27 Oral maintenance therapy consists of active vitamin D analogues (calcitriol 0.25–2 mcg daily or alfacalcidol 0.5–4 mcg daily), with oral elemental calcium (1–3 g/day in divided doses) given separately to avoid impaired absorption.28 Target serum calcium should be maintained in the low-normal range (2.0–2.2 mmol/L) to avoid hypercalciuria and nephrocalcinosis.24

The 2022 British Endocrine Society guidelines and the 2025 revised European Society of Endocrinology Clinical Practice Guideline recommend regular monitoring of serum calcium (every 3–6 months), 24-hour urinary calcium (annually), renal function, and renal ultrasound to screen for nephrolithiasis.29,30 Patients with PHP-Ia or associated features of metabolic

syndrome should receive multidisciplinary management including endocrinology, nutrition, and cardiology input.29

Genetic counselling is an integral component of management for all PHP subtypes given the autosomal dominant inheritance pattern. Cascade genetic testing of first-degree relatives is recommended where available.31 In our patient, genetic testing was recommended but unavailable within the current institutional infrastructure, representing a systems-level gap in care that is likely common across sub-Saharan Africa.

PHP in the Context of Sub-Saharan Africa: Lessons and Gaps

This case carries important lessons for endocrine practice in sub-Saharan Africa. First, the differential diagnosis of tetany and hypocalcaemia in this population must include PHP alongside more commonly considered aetiologies such as vitamin D deficiency, malnutrition-associated hypocalcaemia, and hypoparathyroidism following thyroid surgery. Second, the marked elevation of PTH (176.8 pg/mL) is a critical discriminating finding that, when combined with normal renal function and hyperphosphataemia, effectively excludes all other common causes of hypocalcaemia and should prompt PHP to the forefront of the differential.

Third, the identification of bilateral basal ganglia calcifications on CT brain provides an inexpensive, accessible radiological marker for PHP that can be utilised in settings where genetic testing is unavailable.10 Clinicians in resource-limited settings should maintain a lower threshold for brain CT when PHP is suspected. Fourth, the dyslipidaemia and obesity in our patient — potentially representing partial metabolic syndrome — are recognised associations with PHP and AHO, and should be addressed as part of comprehensive management.

Finally, this case underscores the critical need for a dedicated registry of rare endocrine diseases in Tanzania and the broader East African region. Documenting cases systematically will enable prevalence estimation, improve awareness among clinicians and laboratory scientists, and facilitate access to appropriate investigations and genetic resources through international collaborations such as the European Reference Network on Rare Endocrine Conditions (Endo-ERN).

Comparison with Published Case Reports

A systematic search of PubMed, MEDLINE, and AJOL databases identified no published case reports of PHP from Tanzania or East Africa. Published African case reports are limited to isolated cases from North Africa (Egypt, Morocco), primarily describing PHP Type Ia in paediatric patients with classical AHO features.8 The adult-onset presentation in our patient, occurring at age 43 without AHO phenotype, is atypical even by global standards: a systematic review by Mantovani et al. (2018) reported a mean diagnostic age of 12.5 years for PHP-Ia and 31.7 years for PHP-Ib.11 Our patient's age at diagnosis is therefore most consistent with PHP-Ib or PHP-Ic/II, though this cannot be confirmed without molecular testing.

The presenting complaint of jaw stiffness (trismus-like), which initially prompted referral to the general medicine team and created early diagnostic confusion with tetanus, has been described in severe hypocalcaemia of any aetiology but is particularly prominent in PHP due to the chronicity and severity of calcium depletion.32 The five-month diagnostic delay, during which the patient was treated empirically with vitamin B-complex and pregabalin at the primary healthcare level

without calcium or PTH measurement, is consistent with internationally documented diagnostic delays in PHP averaging 3.5 to 11 years.17

LEARNING POINTS / KEY MESSAGES

1. PHP should be considered in any patient presenting with symptomatic hypocalcaemia, tetany, or paraesthesia, even in the absence of classical AHO features (brachydactyly, short stature, round face, subcutaneous ossifications).

2. The diagnostic key is the biochemical triad: hypocalcaemia + hyperphosphataemia + markedly elevated PTH with normal renal function. This combination effectively excludes hypoparathyroidism, secondary hyperparathyroidism due to renal failure, and vitamin D deficiency as primary diagnoses.

3. Bilateral basal ganglia calcification on CT brain is a hallmark radiological finding in PHP that can support the diagnosis in resource-limited settings where genetic testing is unavailable.

4. Acute management with intravenous calcium gluconate provides rapid symptomatic relief; long-term management with calcitriol and oral calcium is effective and well-tolerated.

5. Genetic counselling should be offered to all PHP patients and their families given the autosomal dominant inheritance pattern with parental imprinting effects.

6. Epidemiological data on PHP in sub-Saharan Africa are absent; this case highlights the need for rare disease registries and improved diagnostic capacity in the region.

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PATIENT CONSENT

Written informed consent was obtained from the patient for publication of this case report and accompanying clinical photographs. Patient identifiers have been anonymised in accordance with BMJ Case Reports editorial policy.

CONTRIBUTORS

JM and SG collected clinical data and drafted the manuscript. DR performed the literature review and contributed to manuscript revision. All authors critically revised the manuscript and approved the final version for submission.

COMPETING INTERESTS

None declared.

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